

Proposed FYDP Ideas for DE-44 (DMTS)

Faculty Name	Proposed Project(s)	Co Supervisor - if Any	No of Students	Relevant SDGs	Funding (Yes/No/partial) / Comments
Dr Danish Hussain	VertiGrow: Edge AI-enabled vertical hydroponic/aeroponic automated agriculture system for Sustainable Smart Farming	Dr Anas, Dr. Basharat Ullah, will work in collaboration with industry	3	9	Maybe
Dr Danish Hussain	AgriDry: AIoT based Apricot dryer	Dr. Basharat Ullah, will work in collaboration with the industry and agricultural division NARC	3	9	No
Dr Danish Hussain	IntelliRover: Design and Development of Intelligent all terrain Mini-Rover for surveillance	Dr. Basharat Ullah, will work in collaboration with industry	3	9	No
Dr Danish Hussain	AI enabled Miniaturized modular agribot for smart farming	Dr Anas, will work in collaboration with industry and Engineering division NARC	3	9	No
Dr Danish Hussain	Development of 3-Axis delta manipulator	Dr. Basharat Ullah	3	9	No
Dr. Zaki Uddin	Design and Development of Solar-Wind Hybrid System for Small-scale Irrigation	Dr. Anjum Naeem Malik	3	7 & 8	Partial
Dr. Zaki Uddin	Exploring the Depths: Advancing Robotics For Precise Pipeline Inspection	Dr. Hammad	3	8	No
Dr. Zaki Uddin	Wireless Power Transfer System For Electric Vehicle Applications	Dr. Basharat Ullah	3	7 & 8	No
Dr. Zaki Uddin	Detection of Electricity Theft Cyber Attacks in Power System			8	
Dr. Amir Hamza	Wheel2Go: Convertible Wheelchair Innovation (Innovating a wheelchair that can be converted into a motorized vehicle, offering greater independence and convenience.)	Dr. Ayesha Zeb	3	10	Partial (bicycle would be available along with electric bicycle conversion kit)
Dr Amir Hamza	Development of a novel vision-based tactile sensor for grasping Applications	Dr Ayesha Zeb, Lt Col Zeeshan Asfar	2-4		Hardware Funding
Dr Amir Hamza	Smart/Hybrid Vertical Axis Wind Turbine for Urban Environment	Dr Hassan Elahi, Dr Anas bin Aqeel	3-4		Fully Funded
Dr Amir Hamza	Multi DOF Robotic Manipulator for STEM Applications	Dr Nasir Rashid	2-4		
Dr Amir Hamza	Real-Time Adaptive Trajectory Generation for Robotic Manipulators:	Lec Usman Asad, A/P Kanwal Naveed	2-4		Dobot CR 5 Available
Dr. Basharat Ullah	Design and Analysis of Linear Oscillating Actuator for Compressors		3	8	No
Dr. Basharat Ullah	Finite Control Set Model Predictive Control of DC-DC Power Electronics Converters	Dr. Zaki Ud Din	3	8	No

Dr. Basharat Ullah	Design and Analysis of Flux Switching Machine for Wind Energy Application		3	7 & 8	No
Dr. Zohaib Riaz	Secure Transit: A smart wearable tracking System for untrusted transport systems				
Dr. Zohaib Riaz	LLM driven UR5 control for high level tasks using Agentic AI	Dr. Ayesha Zeb, Lec Usman Asad	2	9	UR5 available
Dr. Zohaib Riaz	SolarSweep: Smart, Semi-Autonomous Solar Panel Cleaning Robot	Dr. Osama Ali, Dr. Anas	4	8 & 12	
Dr. Zohaib Riaz	EdgePost: A Low-cost Autonomous Mail Delivery Robot with Compute Offloading	Dr. Osama Ali	3		
Dr Zohaib Riaz	Self levelling Delivery Aerial Vehicle	Dr Anas Bin Aqeel, Dr Anjum			
Dr. Zohaib Riaz	PosturePilot: A Vision-Based Smart, Ergonomic Hands-Free Computing Interface	Dr. Osama Ali	2		
Dr. Ayesha Zeb	Intelligent prosthetic hand using sensor fusion and soft robotics	Dr. Umar Shahbaz Khan	2-3	10	DE-43 prosthetic hand along with Raspberry Pi 5 will be available
Dr. Ayesha Zeb	CleanBot: Autonomous Indoor Cleaning Robot	Dr. Kunwar Faraz, Dr. Zohaib Riaz	3	9	Partial (DE-43 CleanBot with some equipment)
Dr. Ayesha Zeb	SceneScribe - AI Enabled Intelligent Glasses for Assistance of Visually Impaired	Dr. Umar Shahbaz Khan Lec. Usman Asad	2	10	Yes
Dr. Hammad Ur Rahman	Design and development of Ventilator for sleep apnea patients	Dr. Anjum Naeem Malik	3	3 & 9	No
Dr. Hammad Ur Rahman	Robotic Ankle-Foot Prosthesis	Dr Hassan Elahi	3	9	No
Dr. Hammad Ur Rahman	Smart Solar Powered Water Filtering System	Dr. Zaki Uddin	3	9	No
Dr Hassan Elahi	AI Enabled diagnostic system	Dr Anjum. Dr Hammad	3	3	No
Dr Hassan Elahi	Design and development of AI Enabled Piezoelectric Ultrasound machine Probe (Biomedical)	Dr Hamid Jabbar. Dr Marco Eugeni	3	3	Partial
Dr Hassan Elahi	Design and development of AI Enabled Structural Health Monitoring Robotic System	Dr Hamid Jabbar. Dr Marco Eugeni	3	9	Partial
Dr Hassan Elahi	Design and development of Piezoelectric based electromagnetic clutches for impact resistant robots	Dr Amir Hmaza. LE Hamza Sohail	3	9	Partial
Dr Hassan Elahi	Design and development of sensors for rehabilitation	Dr Anjum. Dr Hammad	3	3	Partial
Dr. Anjum Naeem	Development of an automated Coagulation Analyzer	Dr. Hammad/Dr. Hassan	3	9	No
Dr. Anjum Naeem	Automated Physical Therapy Robot	Dr. Hassan			
Dr. Anjum Naeem	Design and development of a Highly Accurate Control System for Position and Rate Table	Dr. Zaki Uddin	3	9	Partial
Dr. Anjum Naeem	AI-Based Retinal Scanner for Diabetic Retinopathy Detection	Dr. Tahir Habib			
Dr. Anjum Naeem	Smart Baby Incubator with Vital Sign Monitoring	Dr. Anas			

Dr Hamid Jabbar	Hydrogen Industry stainless steel Pipe Monitoring Robot using Ultrasound, Camera and AI. Existing robotic Vehicles will be used. The Mechanism of Ultrasonic NDT sensor attachment and movement to be developed. Camera interfacing, analysis of image and NDT to be performed for health monitoring.	Dr Zohaib, Professor from Kuwait University, Dr Hassan Elahi	3	9,	No
Dr Hamid Jabbar	PCB Design and Manufacturing- Pakistan OpenSource Robotic/unmanned vehicle Development Board (Containing Processor and related circuit)	LE Umar Aslam, RESOLVE SINES	2	9	May Be
Dr. Umar Shahbaz Khan	DEVELOPMENT AND PRODUCTION OF AN AGILE QUADRUPED ROBOT	Dr. Kunwar Faraz	4	9	Yes (20,000 per month 1 x student will be hired on this salary he will share with the others)
A/P Kanwal Naveed	IndusSafe: Smart helmet for industrial safety			8,9	No
A/P Kanwal Naveed	Autovendi: Autonomous Food Delivery Robot	Usman Asad, Dr Anas		8,9	No
A/P Kanwal Naveed	AutiBot – A friendly robotic assistant for autistic children			8,9	No
A/P Kanwal Naveed	AutoTrolley: Smart Autonomous Cart System for university campus transport			8	No
A/P Kanwal Naveed	An outdoor Autonomous Waste Collection Robot with recycling ability				No
Lec. Usman Asad	HARP: Humanoid Assistive Robotic Platform	A/P Kanwal Naveed	3	8	No
Lec. Usman Asad	Robotic Kitchen Assistant		4	3	No
Lec. Usman Asad	Augmented Reality (AR) based operator assistance		2	8	No
Lec. Usman Asad	Robot training using human demonstrations	Dr. Ayesha Zeb	3	9	No
Dr. Muhammad Osama Ali	Assembly line for filling of viscous fluids using PLC and Conveyor Belts				
Dr. Muhammad Osama Ali	Remote monitoring of Bee Hives using Computer Vision for Enhanced Honey Production				
Dr. Muhammad Osama Ali	Smart LLMs based Personalized Smart Care system for tunnel farming systems	Lec Usman Asad, Dr. Ayesha Zeb	2-3		De-43 tunnel farming system
Dr Anas Bin Aqeel	Design and development of soccer ball launching machine.	Dr Hamza		3,4	
Dr Anas Bin Aqeel	Design and development of a prototype fog collector bio-inspired by desert beetle.	Dr Anjum, Dr Hammad		2,3,4,6	
Dr Anas Bin Aqeel	Design and development of glass cleaning robot.	Dr Hassan		7,9	
Dr Anas Bin Aqeel	Self Levelling Mobile Robot	Dr Uzair, Dr Zohaib		4,9	
Dr Anas Bin Aqeel	Mobile Robot for Smart Restaurants	Dr Zaki, Dr Uzair		2,4,9	
Dr Anas Bin Aqeel	MATE ROV: Underwater Cleaning Robot - Water for sustainable development	Dr Hassan			
Dr Anas Bin Aqeel	Bioinspired footed soft robot				
Dr Tahir Habib Nawaz	Road Condition Assessment in Videos				
Dr Tahir Habib Nawaz	Visual-LiDAR SLAM for Indoor Robot Navigation				
Dr Tahir Habib Nawaz	Autonomous Drone-Based 3D Mapping				
Dr Tahir Habib Nawaz	Abnormal Activity Detection in Videos				
Dr Tahir Habib Nawaz	AI-Powered Parking Lot Monitoring				